

Executive Summary: Guidelines on Use of Corticosteroids in Critically Ill Patients With Sepsis, Acute Respiratory Distress Syndrome, and Community-Acquired Pneumonia Focused Update 2024

Critical illness-related corticosteroid insufficiency (CIRCI) is a state of systemic inflammation with associated dysregulation of the hypothalamus–pituitary–adrenal axis, altered cortisol metabolism, and tissue glucocorticoid resistance (1) that is common in acutely ill patients requiring hospitalization. A multispecialty task force of international experts in critical care medicine and endocrinology from the membership of the Society of Critical Care Medicine and European Society of Intensive Care Medicine previously issued guidelines on the diagnosis of and management of CIRCI across a number of clinical conditions, first in 2008 (2) and then in 2017 (3, 4). Since then, multiple new trials examining the use of corticosteroids in the acutely ill have been published, creating a need to update recommendations inclusive of new evidence. This executive summary provides an update on the previous guidelines on CIRCI, with a focus on sepsis and septic shock, acute respiratory distress syndrome (ARDS), and community-acquired pneumonia (CAP), which were prioritized as the most common diagnoses in which corticosteroids are considered and those with sufficient new data that reevaluation was warranted.

After development of five focused Population, Intervention, Control, and Outcomes questions for this focused update, the panel conducted systematic reviews to identify the best available evidence addressing each question. We evaluated the certainty of evidence using the Grading of Recommendations Assessment, Development, and Evaluation approach and formulated recommendation using the evidence-to-decision framework. The strength of each recommendation was designated as strong (signified by “we recommend”) or conditional (signified by “we suggest”). The panel sought to provide recommendations in both adult and pediatric patient populations, as appropriate, based on available evidence. A summary of recommendations is provided in **Table 1**. A full description of the recommendations is provided in the complete guidelines document.

Corticosteroids in Sepsis and Septic Shock

Recommendation.

- 1A) We suggest administering corticosteroids to adult patients with septic shock (conditional recommendation, low certainty).
- 1B) We recommend against administration of high dose/short duration corticosteroids (defined as > 400 mg/d of hydrocortisone equivalent for < 3 d) for adult patients with septic shock (strong recommendation, moderate certainty).

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TABLE 1.
Summary of Recommendations

Recommendations	Recommendation Strength, Quality of Evidence
Septic shock	
1A. We “suggest” administering corticosteroids to adult patients with septic shock	Conditional recommendation, low certainty evidence
1B. We “recommend against” administration of high dose/short duration corticosteroids (> 400 mg/d hydrocortisone equivalent for less than 3 d) for adult patients with septic shock (strong recommendation, low certainty)	Strong recommendation, moderate certainty evidence
Acute respiratory distress syndrome	
2A. We “suggest” administering corticosteroids to adult hospitalized patients with acute respiratory distress syndrome	Conditional recommendation, moderate certainty evidence
Community-acquired bacterial pneumonia	
3A. We “recommend” administering corticosteroids to adult patients hospitalized with severe bacterial community acquired pneumonia	Strong recommendation, moderate certainty evidence

Remark: We make no recommendation for corticosteroid use in pediatric patients with sepsis.

Rationale. Most studies included patients with septic shock and showed small to moderate desirable effects. Although there was only a small reduction in mortality with low to moderate certainty evidence, there was a larger reduction in shock reversal and organ dysfunction with high certainty evidence (3, 5–10). The panel felt that given the high prevalence of septic shock worldwide, even a small reduction in mortality can have a large effect and that the reduction in shock reversal and organ dysfunction can have important implications in resource utilization. Undesirable effects were felt to be uncertain but anticipated to be small. Further, the intervention was deemed feasible, equitable and acceptable to healthcare providers. The panel did not make a specific recommendation regarding corticosteroid regimen, but recommended against use of high dose, short duration corticosteroids given the risk of adverse effects. Although the panel did not make a specific recommendation on sepsis without shock, if patients present with sepsis and severe CAP or sepsis with ARDS, we suggest administering corticosteroids. Finally, the recommendation for corticosteroid use in sepsis has uncertain generalizability to the pediatric population due to limited available studies.

Corticosteroids in Acute Respiratory Distress Syndrome

Recommendations. 2A) We “suggest” administering corticosteroids to adult critically ill patients

with ARDS (conditional recommendation, moderate certainty).

Remark: We make no recommendation for corticosteroid use in pediatric patients with ARDS.

Rationale. Corticosteroids provide moderate desirable effects, driven primarily by moderate certainty evidence that it reduces hospital mortality (3, 11–15). This effect is more pronounced when corticosteroids are given for more than 7 days. However, the ideal corticosteroid dose, timing, and type remains unknown and left up to clinician comfort and other considerations. This is in contrast to the previous recommendation in 2017 that suggested giving methylprednisolone 1 mg/kg within 14 days of diagnosis of ARDS (3). Undesirable effects of corticosteroids and their cost-effectiveness remain unknown. However, corticosteroid use was deemed feasible and acceptable to healthcare providers. Overall, the panel felt that the benefits of corticosteroid use outweighed its risks. The recommendation has uncertain generalizability to pediatric patients as there were no randomized controlled trials in this population.

Corticosteroids in Community-Acquired Pneumonia

Recommendations. 3A) We “recommend” administering corticosteroids for adult patients hospitalized with severe bacterial community-acquired pneumonia (strong recommendation, moderate certainty).

Remark: We make no recommendation for corticosteroid use in pediatric patients with CAP.

Rationale. Corticosteroids provide large desirable effects in severe CAP with moderate certainty evidence indicating a decrease in hospital mortality and need for invasive mechanical ventilation (4, 16, 17). The same mortality benefit is not seen in patients with less severe CAP (4, 18, 19). Undesirable effects, while unknown as with ARDS and sepsis, are anticipated to be small. Use of corticosteroids in CAP was felt to be feasible, acceptable and may be associated with cost savings (20). As with sepsis and ARDS, the panel was unable to make recommendations on use of corticosteroids for CAP in pediatric patients due to lack of available literature.

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REFERENCES

1. Annane D, Pastores SM, Arlt W, et al: Critical illness-related corticosteroid insufficiency (CIRCI): A narrative review from a Multispecialty Task Force of the Society of Critical Care Medicine (SCCM) and the European Society of Intensive Care Medicine (ESICM). *Intensive Care Med* 2017; 43:1781–1792
2. Marik PE, Pastores SM, Annane D, et al; American College of Critical Care Medicine: Recommendations for the diagnosis and management of corticosteroid insufficiency in critically ill adult patients: Consensus statements from an international task force by the American College of Critical Care Medicine. *Crit Care Med* 2008; 36:1937–1949
3. Annane D, Pastores SM, Rochwerf B, et al: Guidelines for the diagnosis and management of critical illness-related corticosteroid insufficiency (CIRCI) in critically ill patients (Part I). *Crit Care Med* 2017; 45:2078–2088
4. Pastores SM, Annane D, Rochwerf B; Corticosteroid Guideline Task Force of SCCM and ESICM: Guidelines for the diagnosis and management of critical illness-related corticosteroid insufficiency (CIRCI) in critically ill patients (Part II): Society of Critical Care Medicine (SCCM) and European Society of Intensive Care Medicine (ESICM) 2017. *Crit Care Med* 2018; 46:146–148
5. Annane D, Renault A, Brun-Buisson C, et al; CRICS-TRIGGERSEP Network: Hydrocortisone plus fludrocortisone for adults with septic shock. *N Engl J Med* 2018; 378:809–818
6. Agarwal M, Dhar M, Agarwal D, et al: Early initiation of low-dose hydrocortisone therapy for septic shock in geriatric patients: A randomized control trial. *J Assoc Physicians India* 2022; 70:11–12
7. Tongyoo S, Permpikul C: Effect of low dose corticosteroid in septic shock resuscitation: Subgroup analysis result of a randomized controlled trial. *Intensive Care Med Exp* 2018; 6(Suppl 1):P4
8. Talebi Doluee M, Salehi M, Mahmoudi Gharaee A, et al: The effect of physiologic dose of intravenous hydrocortisone in patients with refractory septic shock: A randomized control trial. *J Emerg Pract Trauma* 2018; 4:29–33
9. Birudaraju D, Hamal S, Tayek JA: Solumedrol treatment for severe sepsis in humans with a blunted adrenocorticotrophic hormone-cortisol response: A prospective randomized double-blind placebo-controlled pilot clinical trial. *J Intensive Care Med* 2022; 37:693–697
10. Venkatesh B, Finfer S, Cohen J, et al; ADRENAL Trial Investigators and the Australian–New Zealand Intensive Care Society Clinical Trials Group: Adjunctive glucocorticoid therapy in patients with septic shock. *N Engl J Med* 2018; 378:797–808
11. Sterne JAC, Murthy S, Diaz JV, et al; WHO Rapid Evidence Appraisal for COVID-19 Therapies (REACT) Working Group: Association between administration of systemic corticosteroids and mortality among critically ill patients with COVID-19: A meta-analysis. *JAMA* 2020; 324:1330–1341
12. Tomazini BM, Maia IS, Cavalcanti AB, et al; COALITION COVID-19 Brazil III Investigators: Effect of dexamethasone on days alive and ventilator-free in patients with moderate or severe acute respiratory distress syndrome and COVID-19: The CoDEX randomized clinical trial. *JAMA* 2020; 324:1307–1316
13. Jeronimo CMP, Farias MEL, Val FFA, et al; Metcovid Team: Methylprednisolone as adjunctive therapy for patients hospitalized with COVID-19 (Metcovid): A randomised, double-blind, Phase IIb, placebo-controlled trial. *Clin Infect Dis* 2021; 72:e373–e381
14. Horby P, Lim WH, Emberson JR, et al: Dexamethasone in hospitalized patients with COVID-19. *N Engl J Med* 2021; 384:693–704
15. Villar J, Ferrando C, Martínez D, et al; dexamethasone in ARDS network: Dexamethasone treatment for the acute respiratory distress syndrome: A multicentre, randomised controlled trial. *Lancet Respir Med* 2020; 8:267–276
16. Meduri GU, Shih MC, Bridges L, et al; ESCAPE Study Group: Low-dose methylprednisolone treatment in critically ill patients with severe community-acquired pneumonia. *Intensive Care Med* 2022; 48:1009–1023
17. Dequin P-F, Meziani F, Quenot J-P, et al; CRICS-TRIGGERSEP Network: Hydrocortisone in severe community-acquired pneumonia. *N Engl J Med* 2023; 388:1931–1941
18. Lloyd M, Karahalios A, Janus E, et al; Improving Evidence-Based Treatment Gaps and Outcomes in Community-Acquired Pneumonia (IMPROVE-GAP) Implementation Team at Western Health: Effectiveness of a bundled intervention including adjunctive corticosteroids on outcomes of hospitalized patients with community-acquired pneumonia: A stepped-wedge randomized clinical trial. *JAMA Intern Med* 2019; 179:1052–1060
19. Wittermans E, Vestjens SMT, Spoorenberg SMC, et al; Santeon-CAP Study Group: Adjunctive treatment with oral dexamethasone in non-ICU patients hospitalised with community-acquired pneumonia: A randomised clinical trial. *Eur Respir J* 2021; 58:2002535
20. Pliakos EE, Andreatos N, Tansarli GS, et al: The cost-effectiveness of corticosteroids for the treatment of community-acquired pneumonia. *Chest* 2019; 155:787–794